

NOTICE EXPLAIN MATCH UPLIFT NOTICE EXPLAIN MATCH UPLIFT

Nemu

Notice Explain Match Uplift.

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Topic

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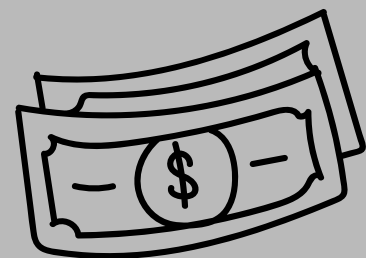
From Invisible Markets to Spend Capture

Uncovering Demand and Optimising Incentives

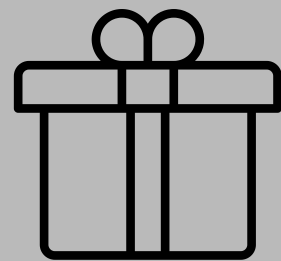
Revealing how localized physical spending obscures true market potential, and introducing dynamic behavioral interventions to recapture lost revenue efficiently.

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Dual Problems



Amex loses massive overseas revenue to uncaptured local transactions due to a data blindspot.



We waste profit margins deploying broad, untargeted rewards instead of precise interventions.

Illustrating the Challenge

Sarah books a luxury trip to Bangkok, including flights and accommodations, using her American Express card.

Once in Bangkok, Sarah spends at local markets and merchants that do not accept American Express.

The Critical Miss: Because of this blindspot, Amex completely failed to identify Bangkok as a massive potential market.

The Data Trap

What Amex See

Flight and Hotel

\$1200

Her Journey ends in our database

The Reality

Total Physical Trip Spend

\$3500

The \$2,300 gap is the Invisible Market.

The Margin Paradox

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We Subsidize Travel

Amex finances the initial \$1,200 trip with expensive points and perks, capturing the lowest-margin segment of her journey.



Competitors Win Local

Competitors easily process the remaining \$2,300 in high-frequency local spend with virtually no reward overhead or margin cost.



The Missing Link

Why aren't we capturing the full \$3,500? Because we lack the mechanism to influence her payment choice at the physical point of sale.

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Measuring the Unseen

Only Amex sees the entire trip because we book the travel.

If Sarah usually spends \$400 on food at home, but we only record \$30 in Bangkok, the missing \$370 is our signal.

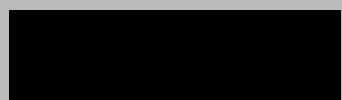
She didn't stop eating. She just paid another way. This allows us to map lost revenue globally.

The Next Challenge

How do we capture that missing \$2,300?

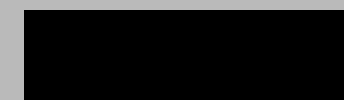
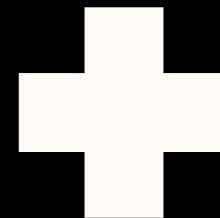
Broad rewards waste margin. We need dynamic behavioral interventions.

The Flaw of Broad Rewards



The "Sure Thing":

Sending a generic "5% Cashback" to a loyal customer wastes profit margin on someone who already intended to use Amex.



The "Friction Point":

Missing the exact moment a user switches to cash due to a 2% local fee. A micro-targeted nudge right then saves the transaction.

Dynamic Interventions

Too Expensive

Broad Subsidies

Indiscriminate offers destroy margin on users who required no persuasion to use our card.

The Optimal Choice

Micro-Nudges

A precisely timed intervention offsets local friction, acquiring the transaction while preserving profit.

Lost Revenue

No Intervention

Without the right nudge at a friction point, the customer defaults to cash. The invisible market grows.

Turn cross-border opportunity into measurable billed business.

REVEAL UNMET DEMAND

Estimate where Card Members may be spending during a trip without using Amex.

ROUTE THE RIGHT RESPONSE

After identifying why spend was missed, NEMU recommends the action most likely to recover it.

If Amex acceptance is limited, NEMU identifies the merchants that should be prioritised for Amex acceptance.

If Amex is accepted but the Card Member prefers another card, NEMU recommends the smallest offer likely to change their behaviour.

If the spend was likely paid in cash or there was no genuine demand, NEMU recommends taking no action, avoiding unnecessary investment.

MEASURE WHAT ACTUALLY WORKED

Compare every action with a control group and count only the additional billed business created.



NEMU Decision Framework

NEMU gives Amex a new way to:

Notice cross-border spending opportunities that transaction data alone may not reveal.

Explain how much of that opportunity is realistically recoverable.

Match each opportunity with the right intervention, whether merchant acquisition, a targeted offer, or no action.

Uplift incremental billed business and verify return before scaling

Implementation Plan From Insights to Actions

Turning hidden demand into targeted customer engagement, measurable value, and scalable growth.



Ranking Dashboard

Prioritize the top 2–3 corridor-category opportunities each quarter using NEMU recoverable value instead of historical Amex volume.



Merchant Target List

Focus merchant acquisition on high-potential clusters with proven unmet demand, starting with controlled pilot districts.



Incentive Recommendation

Target only persuadable Card Members with the optimal offer, while maintaining holdouts and allowing “send nothing” when appropriate.



“Card Works Here” Map

Increase destination spend by showing travelers where Amex is accepted, using a low-cost, test-and-learn approach.



Leadership Report

Track predicted vs. actual value and intervention costs to determine whether investments should scale.

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The Impact

For Amex

Recaptures pure overseas revenue instead of subsidizing existing behavior.

For Merchants

Optimizes budgets, paying only to overcome friction for highly qualified spenders.

For Card Members

Delivers frictionless, intelligent interventions precisely when needed.

One Solution, Three Outcomes

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Productivity

Automate travel-market analysis, merchant prioritisation, offer selection, and performance measurement.

Protection

Use aggregated data for market planning and require explicit Card Member consent before sending individual offers.

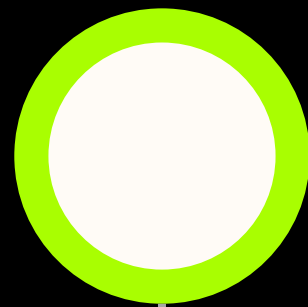
Growth

Increase additional cross-border billed business and improve how merchant and offer budgets are allocated.

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Implementation Plan Roadmap

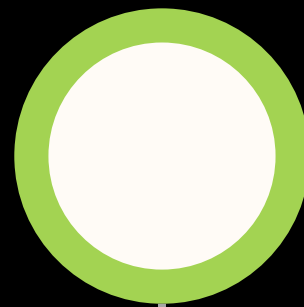
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Phase 1

Estimate & Validate

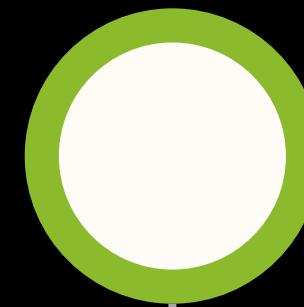
Analyze historical trip transactions to estimate missing spend and validate the results through a sampled post-trip Card Member survey.



Phase 2

Route & Test

Route acceptance-driven opportunities to merchant acquiring teams in target countries and test preference-driven opportunities through opt-in Amex Offers with a no-offer control group.



Phase 3

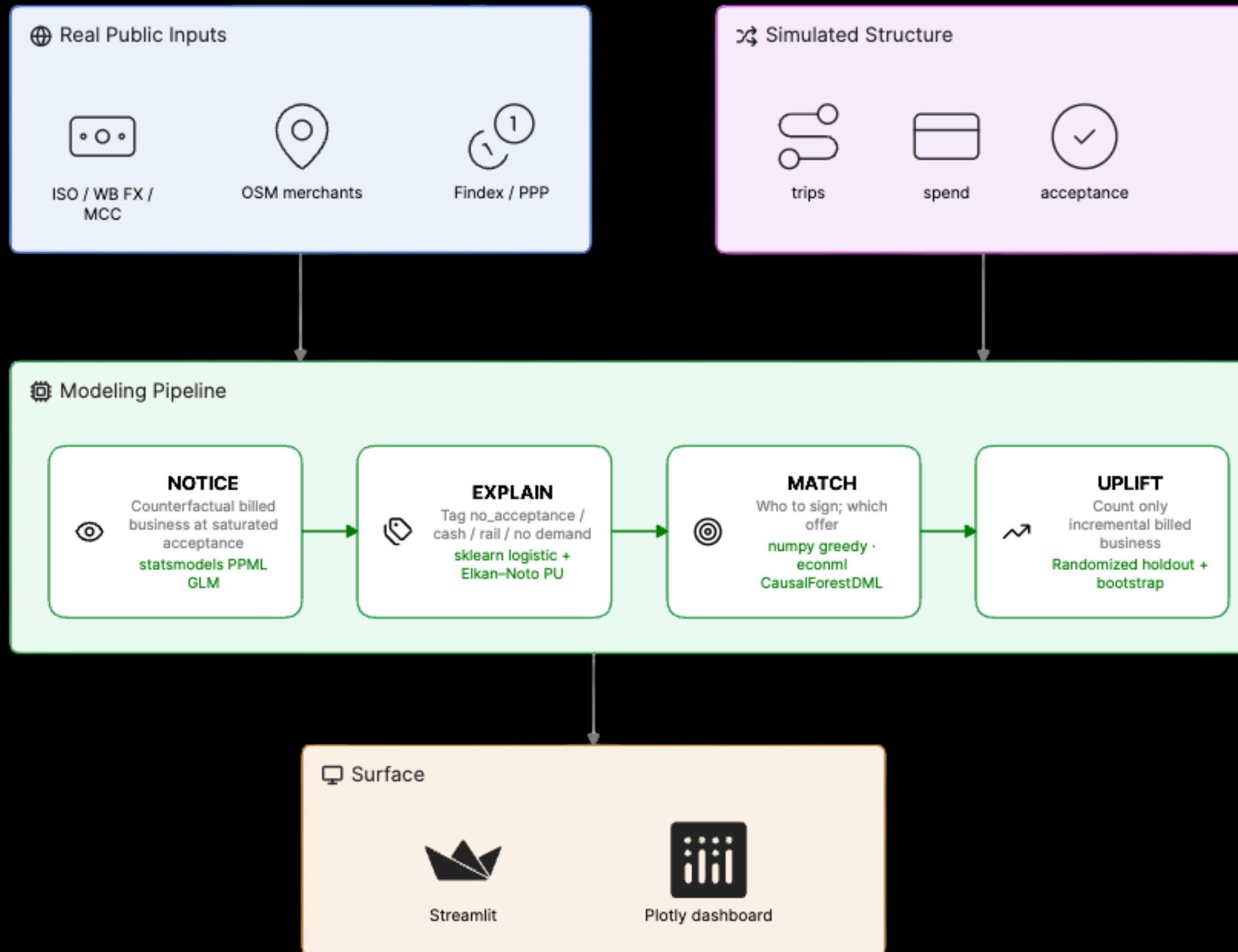
Scale

Expand to additional APAC travel corridors only after realized incremental value consistently exceeds merchant acquisition and incentive costs.

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How We Implement It

Overview



How We Implement It

Real Inputs & Simulated Structure

- **ISO / World Bank FX / MCC**: real currency codes, official FX, real merchant categories. So a ticket looks like a ticket, not a fake column.
- **OSM merchants**: real venue names on the map. Which member lands on which name is simulated; the name itself is public.
- **Findex / PPP**: cash intensity and price levels by country. That is the confound. Vietnam is both sparse and cash-heavy. If we do not pin cash to Findex, Notice will call "cash culture" a coverage hole.

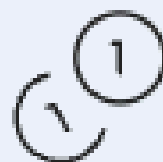
Real Public Inputs



ISO / WB FX /
MCC



OSM merchants

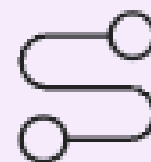


Findex / PPP

Simulated structure is trips, spend, and acceptance density.

That last one is the treatment. It has to be simulated because OSM counts restaurants, not Amex terminals. We designed sparse vs saturated districts inside the same country (Ubud vs SCBD, Khao San vs Sukhumvit) so the gravity coefficient is identified off coverage, not off "Indonesia is poorer than Japan."

Simulated Structure



trips



spend

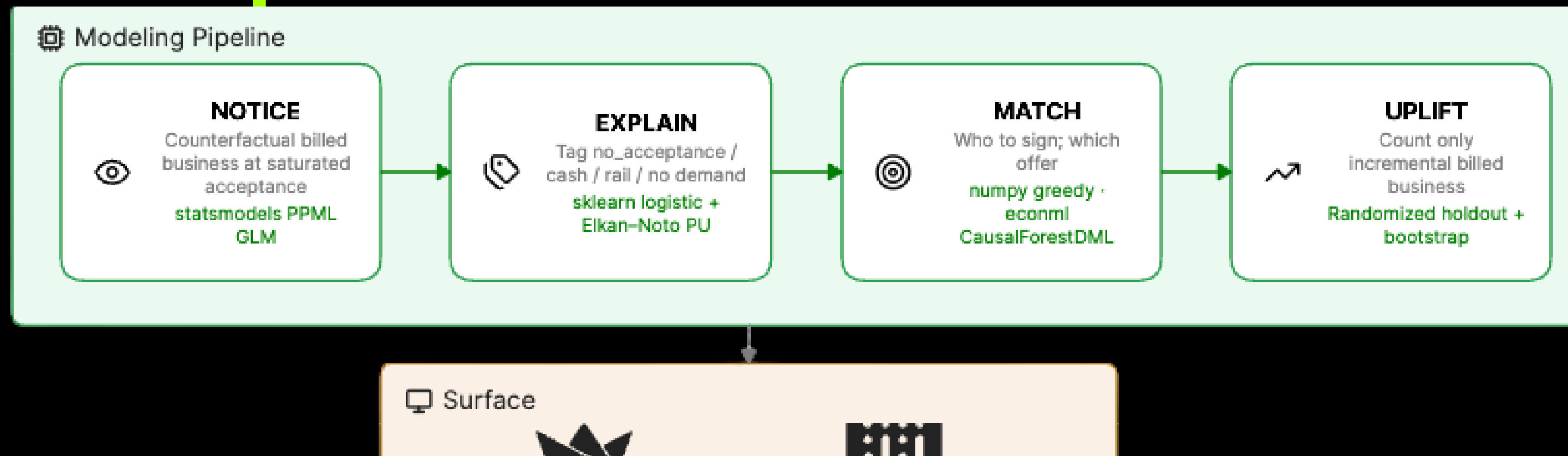


acceptance

Modeling Pipeline

How We Implement It

Model Pipeline

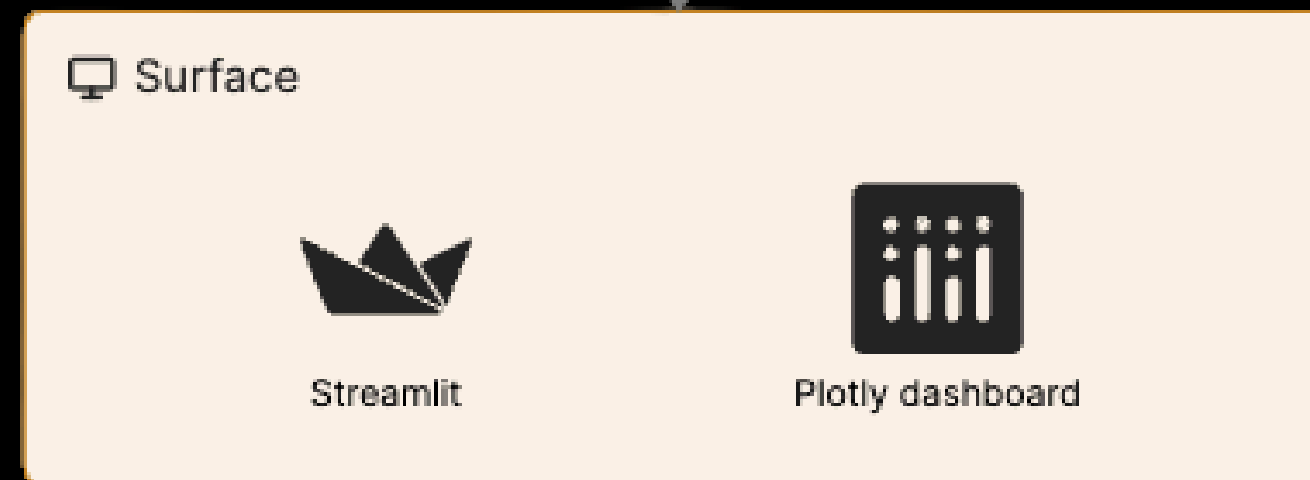


1. **Notice**: statsmodels PPML. Reconstruct the trip × category grid, including zeros. Ask: if acceptance were 0.95, how much billed business? Not log-OLS, because zeros and heteroskedasticity would bias $y|x$.
2. **Explain**: sklearn logistic + Elkan-Noto PU. Rules first (no demand / no acceptance / cash / rail). Declines are the only confirmed “card not taken” labels; silence is unlabeled, not a negative.
3. **Match**: greedy submodular for who to sign (diminishing coverage, not dump the budget into Ubud). CausalForest for which offer (2x_points vs statement credit) maximises net value, not raw uplift.
4. **Uplift**: freeze the policy, 50/50 holdout, bootstrap. Count only incremental billed business. Calibration ~1.01 means we are on scale, not just ranking people.

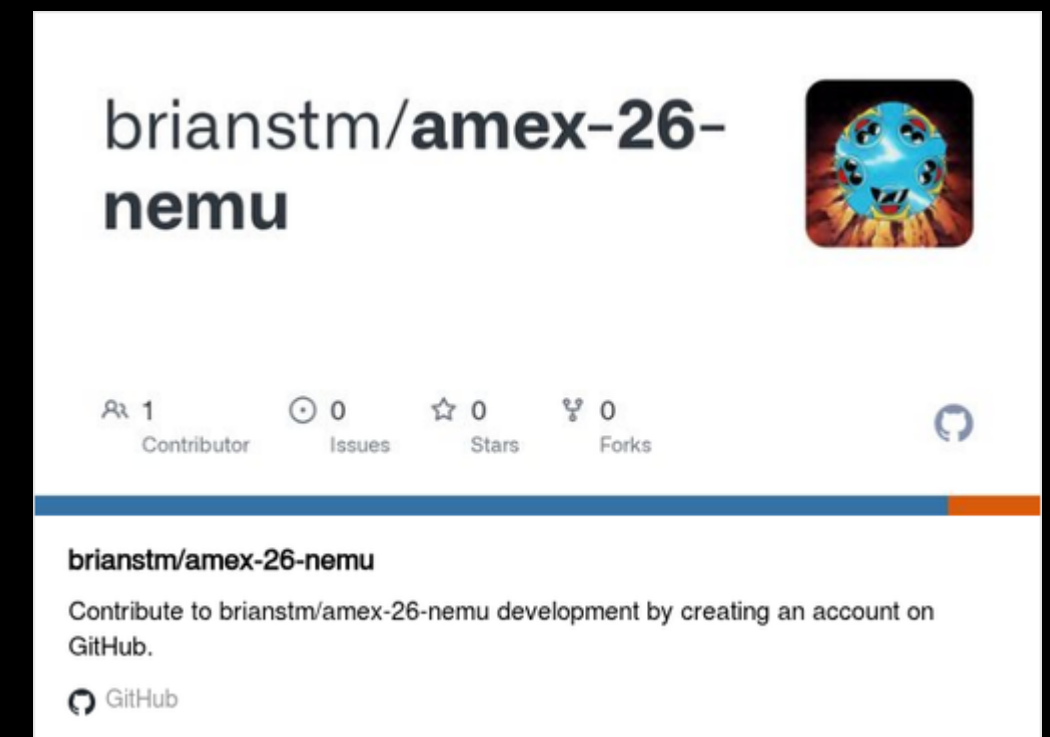
How We Implement It

How we Implemented it.

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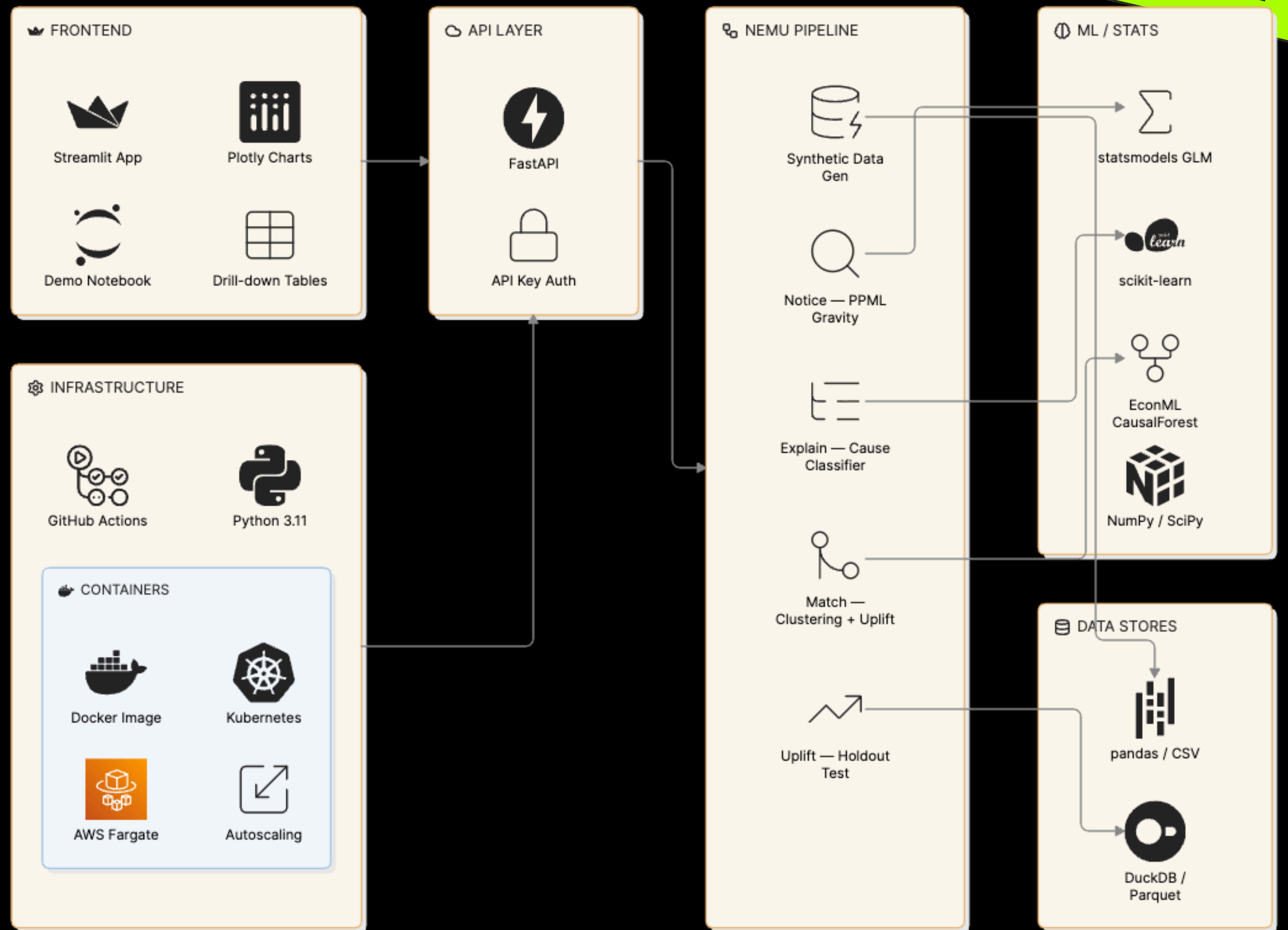
Surface is **Streamlit** + **Plotly** so a BD or finance partner can see the corridor, causes, and holdouts



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Architecture

How an actual implementation would be done



Real (public)	Simulated (so we can score Notice)
 ISO 4217 currency World Bank FX ISO MCC	 Who travelled How much they spent
 OSM / public merchant names (Ichiran, Jumbo Seafood, ThongSmith, etc.)	 Which ticket is assigned to which name
 Findex cash PPP price level	 District acceptance density (the treatment)

- ISO 4217 — currency codes (SGD, THB, JPY, ...) www.iso.org/iso-4217-currency-codes.html
- World Bank FX — official LCU per USD (PA.NUS.FCRF) data.worldbank.org/indicator/PA.NUS.FCRF
- ISO 18245 MCC — merchant category codes (5812 dining, 5311 retail, 4111 transport, 7011 lodging) www.iso.org/standard/33365.html
- OpenStreetMap — merchant / POI names (ODbL) www.openstreetmap.org
- Nominatim — district geocoding nominatim.openstreetmap.org
- Overpass API — amenity / hotel counts near each district overpass-api.de
- Global Findex 2025 — cash / digital-payment share (fin25e2, fin25e2b, g20.made) www.worldbank.org/en/publication/globalfindex
- World Bank PLI / PPP — price level vs Singapore (PA.NUS.GDP.PLI) data.worldbank.org/indicator/PA.NUS.GDP.PLI
- WDI tourist arrivals — trip-volume weights, 2015-2019 (ST.INT.ARVL) data.worldbank.org/indicator/ST.INT.ARVL

Issuer tickets (real currency + OSM merchants)

Spend and which trip the ticket belongs to are simulated. `currency` is ISO 4217, `amount_local` uses World Bank FX, `merchant_name` is an OpenStreetMap POI (or a public transport operator), `mcc` is ISO 18245. This is not an Amex merchant file.

txn_id	dest_country	dest_district	merchant_name	category	mcc	currency	amount_local	amount_usd
X0000000	Japan	Namba	Ichiran Dotonbori	dining	5812	JPY	79519	531.34
X0000001	Indonesia	SCBD	Cork & Screw	dining	5812	IDR	628348	38.13
X0000002	Vietnam	Ba Dinh	Cà Phê Trang	dining	5812	VND	2514001	104.04
X0000003	Thailand	Sukhumvit	Thongsmith	dining	5812	THB	10312.3	313.61
X0000004	Singapore	Orchard	Jumbo Seafood	dining	5812	SGD	911.32	697.02
X0000005	Hong Kong	Mong Kok	Yat Lok	dining	5812	HKD	1612.82	206.85
X0000006	South Korea	Myeongdong	소호	dining	5812	KRW	1090090	766.35
X0000007	Vietnam	Han River	MienThao Coffee	dining	5812	VND	571735	23.66
X0000008	Malaysia	KLCC	Din Tai Fung Suria KLCC	dining	5812	MYR	149.29	34.84
X0000009	France	Le Marais	L'As du Fallafel	dining	5812	EUR	853.16	964.06



\$7.34M

Hidden acceptance leakage



\$6.82M (93%)

Model recovered



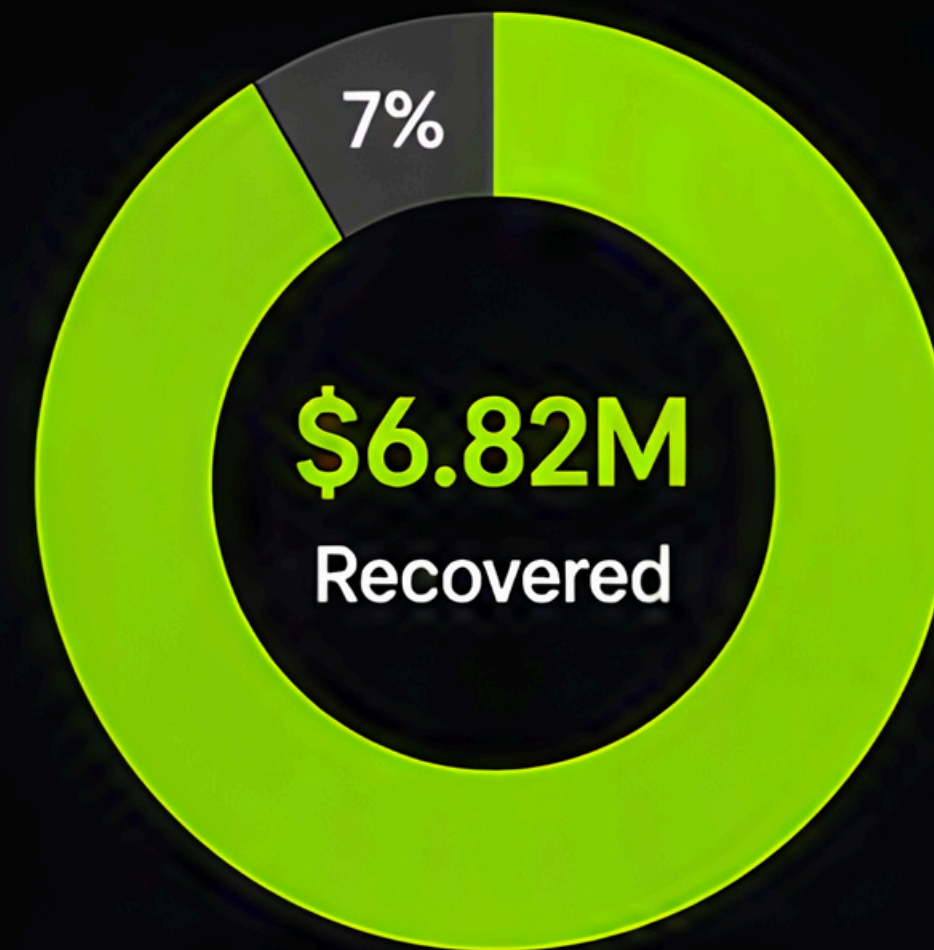
92%

Cause \$ accuracy



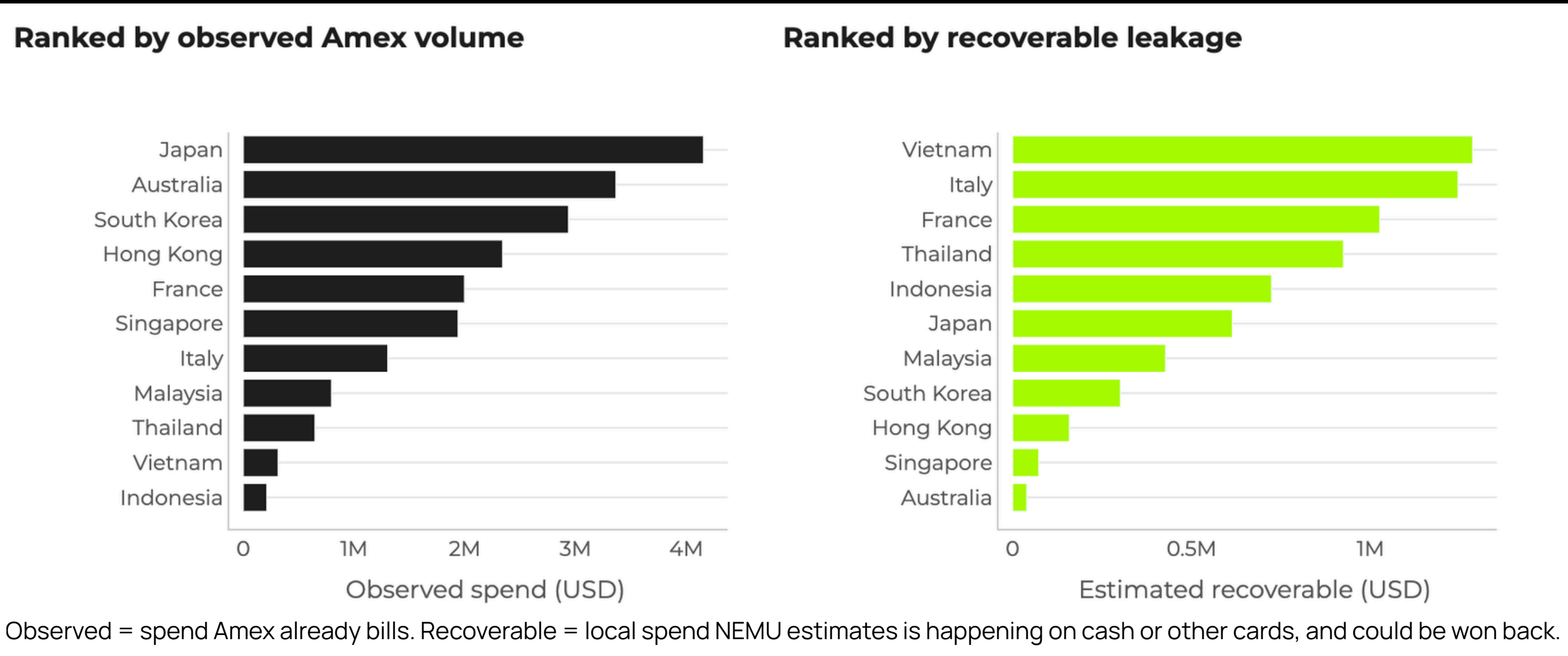
1.01

Holdout calibration



Today's Biggest Markets Aren't Where the Growth Is

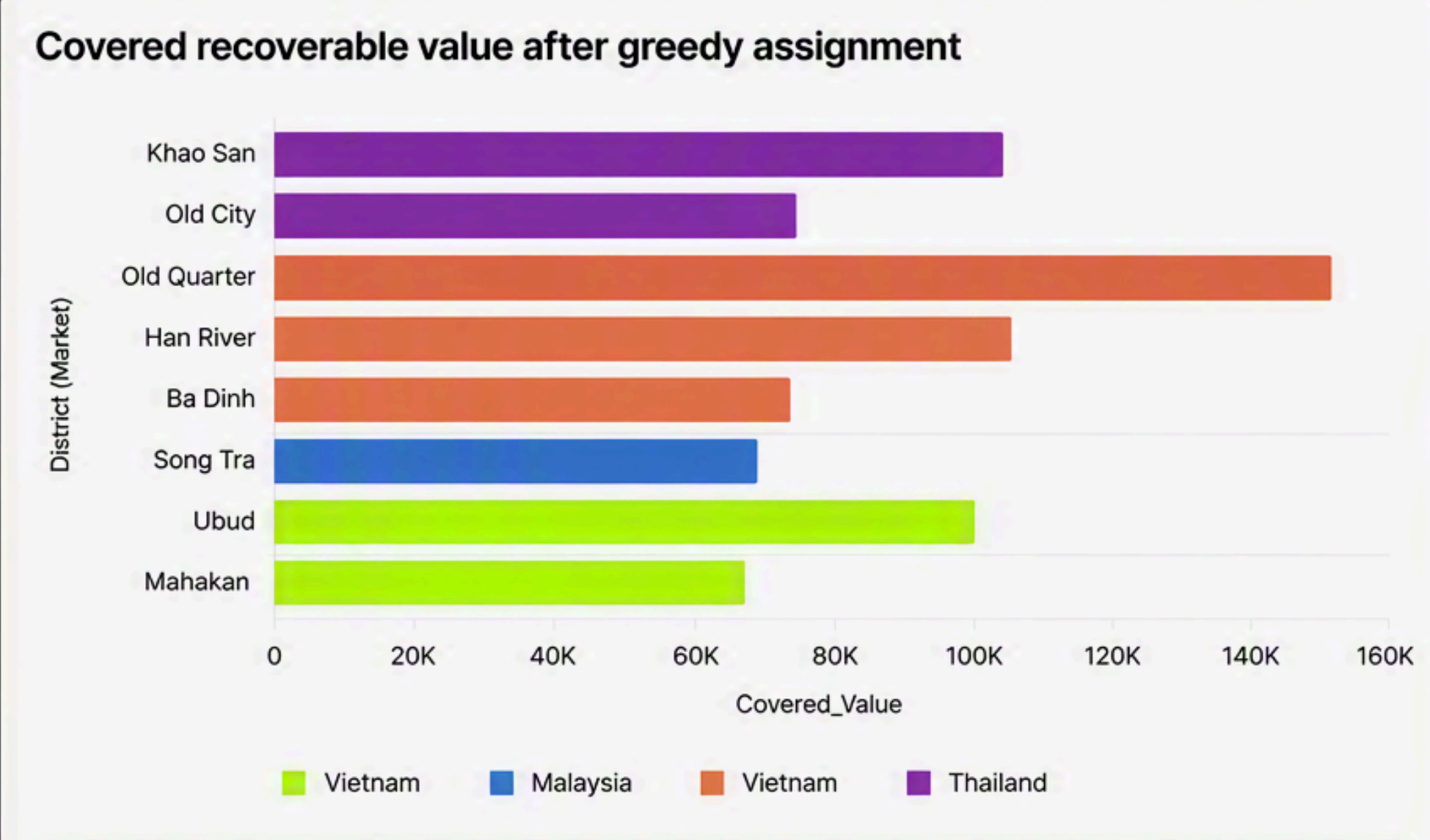
Ranking countries by current Amex billing tells a different story than ranking them by recoverable spend. Vietnam, Italy, and France sit near the bottom today — but hold the largest pools of local spend Amex isn't capturing. Japan and Australia bill the most, but have the least headroom left. Investment should follow recoverable value, not historical volume.



This ranking was tested, not assumed: we hid \$7.34M of known coverage leakage in the data, and the model recovered 93% of it, attributing 92% of the dollars to the correct cause, without ever seeing the answer key. That's what lets us act on the ranking above with confidence.

4 SEA Districts Amex Should Target First

Based on our data simulation and public datasets, Amex should prioritize merchant acquisition in four districts across Vietnam, Thailand, and Indonesia, Old Quarter, Han River, Khao San, and Ubud — where uncaptured local spend is both large and concentrated. Targeting these first is expected to recover USD 145K in currently missed spend and generate USD 243K in net incremental billed business.



Target Merchant District
Old Quarter (VN): 4 merchants
Han River (Vietnam), Khao San (Thailand), Ubud (Indonesia)

Recoverable Value Coverage
USD 145K

Expected Net Incremental Uplift
USD 243K

This shortlist comes from a model we've validated, not just optimized: in holdout testing, predicted uplift (USD 201) came within 1% of the realized result (USD 203) — a 1.01x calibration — giving confidence this selection will hold up at scale.

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Thank You

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In fulfilment of Amex AI Hackathon 2026

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